

FBISE Mathematics SSC-I

Class 9 - 1st Annual Examination

2026

Complete Solutions of Local Area, Hard Area and Overseas Papers

All 45 MCQs explained with correct-answer reasoning and option-by-option elimination

Every part of Sections B and C solved, including all OR alternatives

Stepwise, exam-ready working designed for full-mark presentation

Prepared by Fakhar STEM Sphere

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Accuracy note: Solutions follow the wording visible in the supplied scanned papers. Where a printed question is under-specified or mathematically ambiguous, the issue is stated clearly instead of assuming missing data.

Paper 1 - Version 10081

Section A - MCQs with detailed reasoning

MCQ 1. Evaluate $(1/16)^{-3/2}$.

A. $1/4$	B. 4	C. 16	D. 64
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Correct answer: D. A negative exponent takes the reciprocal: $(1/16)^{-3/2} = 16^{3/2} = (\sqrt{16})^3 = 4^3 = 64$.

Why the other options are wrong: A is $16^{-1/2}$; B is $16^{1/2}$; C is 16^1 . None uses the required exponent $3/2$.

MCQ 2. Write 4500 in scientific notation.

A. 4.5×10^3	B. 45×10^3	C. 4.5×10^2	D. 0.45×10^5
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Correct answer: A. Scientific notation requires $1 \leq \text{coefficient} < 10$. Moving the decimal three places gives 4.5×10^3 .

Why the other options are wrong: B = 45,000; C = 450; D = 45,000, and B/D also fail the required coefficient rule.

MCQ 3. Which one is a binary relation?

A. $\{1,2,3\}$	B. $\{(1,2),(2,3)\}$	C. $\{1 \rightarrow 2 \rightarrow 3\}$	D. $\{1,2\} \times \{3\}$
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Correct answer: B. A binary relation is represented as a set of ordered pairs, so B is the intended answer.

Why the other options are wrong: A is a set of numbers, not ordered pairs; C is not standard set notation. D denotes a Cartesian product rather than listing the relation. Strictly, that product expands to ordered pairs and can itself be a relation, but the board's intended explicit relation is B.

MCQ 4. Find the HCF of $-12x^2y^2$, $6xy^3$, $24x^2y^2$.

A. $-6xy$	B. $-6x^2y^3$	C. $6xy^2$	D. $24x^3y^2$
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Correct answer: C. HCF of coefficients is 6; minimum powers are x^1 and y^2 . Hence HCF = $6xy^2$.

Why the other options are wrong: A omits a common y; B uses powers not present in every term and an unnecessary negative sign; D uses the largest coefficient and excessive x-power.

MCQ 5. $x + 3 > 5$ implies:

A. $x = 0$	B. $x < 2$	C. $x \leq 2$	D. $x > 2$
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Correct answer: D. Subtracting 3 from both sides gives $x > 2$.

Why the other options are wrong: A does not satisfy the inequality; B and C reverse or include values that make $x + 3 \leq 5$.

MCQ 6. In the right triangle, hypotenuse = 4 and the side opposite $\theta = 2$. Find θ .

A. 90°	B. 60°	C. 30°	D. 0°
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Correct answer: C. $\sin \theta = \text{opposite/hypotenuse} = 2/4 = 1/2$, so $\theta = 30^\circ$.

Why the other options are wrong: $\sin 90^\circ = 1$, $\sin 60^\circ = \sqrt{3}/2$ and $\sin 0^\circ = 0$; none equals $1/2$.

MCQ 7. A plane flies on a bearing of 180° . It is moving toward:

A. North	B. South	C. East	D. West
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Correct answer: B. Bearings are measured clockwise from North; 180° points due South.

Why the other options are wrong: North is $000^\circ/360^\circ$, East is 090° , and West is 270° .

MCQ 8. If two triangles are equal, their corresponding sides are:

A. Equal	B. Unequal	C. Unproportional	D. Perpendicular
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Correct answer: A. Equal (congruent) triangles have equal corresponding sides.

Why the other options are wrong: B and C contradict congruence; perpendicular describes an angle relation, not corresponding side lengths.

MCQ 9. $1/\sin \theta$ equals:

A. $\tan \theta$	B. $\cos \theta$	C. $\sec \theta$	D. $\operatorname{cosec} \theta$
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Correct answer: D. Cosecant is the reciprocal of sine: $\operatorname{cosec} \theta = 1/\sin \theta$.

Why the other options are wrong: $\tan \theta = \sin \theta / \cos \theta$; $\cos \theta$ is not a reciprocal; $\sec \theta = 1/\cos \theta$.

MCQ 10. The point where the medians of a triangle intersect is called:

A. Incentre	B. Circumcentre	C. Centroid	D. Orthocentre
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Correct answer: C. The three medians are concurrent at the centroid.

Why the other options are wrong: Incentre: angle bisectors; circumcentre: perpendicular bisectors; orthocentre: altitudes.

MCQ 11. A triangular road sign has all sides equal. Its type is:

A. Right-angled	B. Equilateral	C. Isosceles	D. Scalene
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Correct answer: B. A triangle with all three sides equal is equilateral.

Why the other options are wrong: Right-angled concerns one 90° angle; isosceles has at least two equal sides but the specific three-equal classification is equilateral; scalene has no equal sides.

MCQ 12. The locus of points equidistant from two fixed points is the:

A. Perpendicular bisector	B. Angle bisector	C. Median	D. Altitude
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Correct answer: A. Every point on the perpendicular bisector of a segment is equally distant from its endpoints.

Why the other options are wrong: Angle bisectors concern equal distance from intersecting lines; medians and altitudes are triangle segments and do not give this locus.

MCQ 13. Find the class mark of interval 15-20.

A. 15	B. 17.5	C. 20	D. 35
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Correct answer: B. Class mark = (lower limit + upper limit)/2 = (15 + 20)/2 = 17.5.

Why the other options are wrong: A and C are class limits; D is the sum, not the midpoint.

MCQ 14. The probability of an event cannot be:

A. 0	B. 0.5	C. 1	D. 2
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Correct answer: D. Every probability lies in $0 \leq P(E) \leq 1$, so 2 is impossible.

Why the other options are wrong: 0, 0.5 and 1 are valid probabilities for impossible, uncertain and certain events respectively.

MCQ 15. A boy scored goals in 3 of 9 matches. His relative frequency is:

A. 9	B. 3	C. $1/2$	D. $1/3$
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Correct answer: D. Relative frequency = successes/trials = $3/9 = 1/3$.

Why the other options are wrong: 9 is the number of trials; 3 is the number of successes; $1/2$ would mean half the matches.

Section B - Complete solutions (including every OR part)

Q2(i) Simplify using laws of indices

$$(a) 2^{2/3} \times 2^{1/7} \div 2^{1/2} = 2^{(2/3 + 1/7 - 1/2)} = 2^{(28+6-21)/42} = 2^{13/42}.$$

$$(b) 3^{-1/2} \times 3^{1/2} \div 3^{1/2} = 3^{-1/2+1/2-1/2} = 3^{-1/2} = 1/\sqrt{3}.$$

Answers: (a) $2^{13/42}$; (b) $1/\sqrt{3}$.

OR: Probability from a bag

Total balls = 10; white = 3; black = 7.

$$P(\text{white}) = 3/10. P(\text{black}) = 7/10.$$

$$P(\text{white}) + P(\text{black}) = 3/10 + 7/10 = 1.$$

Answers: 3/10, 7/10 and 1.

Q2(ii) Plane travels 600 km on bearing 300°

Bearing 300° is N60°W, equivalently W30°N. The westward component is adjacent to the 30° angle measured from West.

$$\text{Westward distance} = 600 \cos 30^\circ = 600(\sqrt{3}/2) = 300\sqrt{3} \text{ km} \approx 519.6 \text{ km}.$$

The plane travelled $300\sqrt{3}$ km $\approx$ 519.6 km west.

OR: Compare Richter-scale intensities

$$M = \log_{10}(I/I_0). \text{ Hence } M_C - M_P = \log(I_C/I_P).$$

$$8.2 - 7.6 = 0.6, \text{ so } I_C/I_P = 10^{0.6} \approx 3.981.$$

China's earthquake was about 3.98 times as intense.

Q2(iii) Cartesian product and relation

$$A \times B = \{(1,2), (1,4), (3,2), (3,4)\}.$$

For $R_1 = \{(x,y) | x \in A, y \in B \text{ and } x > y\}$, test each pair. Only $3 > 2$.

$A \times B = \{(1,2), (1,4), (3,2), (3,4)\}$; $R_1 = \{(3,2)\}$.

OR: Line through (1,3) and the intersection of two lines

Solve $2x - 3y + 4 = 0$ and $4x + y - 1 = 0$. From the second, $y = 1 - 4x$.

$$2x - 3(1 - 4x) + 4 = 0 \Rightarrow 14x + 1 = 0 \Rightarrow x = -1/14; y = 9/7.$$

Slope through $(-1/14, 9/7)$ and $(1,3)$: $m = (3 - 9/7)/(1 + 1/14) = 8/5$.

$$y - 3 = (8/5)(x - 1) \Rightarrow 8x - 5y + 7 = 0.$$

Required line: $8x - 5y + 7 = 0$.

Q2(iv) Compound inequality

$$-3x + 11 \geq 5(x + 7) - 32 \geq 3x - 1 \text{ becomes } -3x + 11 \geq 5x + 3 \text{ and } 5x + 3 \geq 3x - 1.$$

$$\text{First: } 8 \geq 8x \Rightarrow x \leq 1. \text{ Second: } 2x \geq -4 \Rightarrow x \geq -2.$$

Intersection: $-2 \leq x \leq 1$. On the number line, use solid dots at -2 and 1 and shade between them.

Solution set: $[-2, 1]$.

OR: Factorize $3x^2 + 16x - 12$

Product $a \times c = 3(-12) = -36$. Numbers 18 and -2 have product -36 and sum 16.

$$3x^2 + 18x - 2x - 12 = 3x(x + 6) - 2(x + 6).$$

$3x^2 + 16x - 12 = (3x - 2)(x + 6)$.

Q2(v) Solve right triangle ABC

Given $\angle C = 90^\circ$, $c = 4\sqrt{2}$ cm is the hypotenuse and $a = 4$ cm.

$$b = \sqrt{(c^2 - a^2)} = \sqrt{(32 - 16)} = 4 \text{ cm}.$$

Since $a=b$, the two acute angles are equal: $A=B=(180^\circ-90^\circ)/2=45^\circ$.

$a=4\text{ cm}$, $b=4\text{ cm}$, $c=4\sqrt{2}\text{ cm}$; $\angle A=45^\circ$, $\angle B=45^\circ$, $\angle C=90^\circ$.

OR: Verify $\sin^2\theta+\cos^2\theta=1$

In a right triangle let opposite= p , adjacent= b and hypotenuse= h . Then $\sin\theta=p/h$ and $\cos\theta=b/h$.

$\sin^2\theta+\cos^2\theta = p^2/h^2+b^2/h^2 = (p^2+b^2)/h^2 = h^2/h^2 = 1$ by Pythagoras.

Verified: $\sin^2\theta + \cos^2\theta = 1$.

Q2(vi) Gradient through A(2,3), B(6,11)

$m=(y_2-y_1)/(x_2-x_1)=(11-3)/(6-2)=8/4=2$.

Gradient = 2.

OR: Verify associativity of union

$P\cup Q=\{0,1,2,3,4,5,6\}$; therefore $(P\cup Q)\cup R=\{0,1,2,3,4,5,6,7,8,9\}$.

$Q\cup R=\{2,3,4,5,6,7,8,9\}$; therefore $P\cup(Q\cup R)=\{0,1,2,3,4,5,6,7,8,9\}$.

Both sides are equal, so $(P\cup Q)\cup R = P\cup(Q\cup R)$.

Q2(vii) Fountain equidistant from P(4,7) and Q(10,1)

Let the fountain be (x,y) . Equate squared distances:

$(x-4)^2+(y-7)^2=(x-10)^2+(y-1)^2$.

After expansion and cancellation: $12x-12y-36=0 \Rightarrow x-y-3=0$.

All possible positions lie on $y=x-3$, the perpendicular bisector of PQ.

OR: Line through (-4,-4) parallel to slope -5

A parallel line has slope -5. Using point-slope form: $y+4=-5(x+4)$.

$y=-5x-24$, or $5x+y+24=0$.

Q2(viii) Similar cylinders

Take base-area ratio large:small = 9:4. Thus $A_L/240=9/4$, so $A_L=540\text{ cm}^2$.

For the large base, $\pi r^2=540 \Rightarrow r=\sqrt{(540/\pi)}\approx 13.11\text{ cm}$; diameter= $2r\approx 26.22\text{ cm}$.

Large cylinder: base area 540 cm^2 , radius $\sqrt{(540/\pi)}\approx 13.11\text{ cm}$, diameter $\approx 26.22\text{ cm}$.

The wording implies the requested radius and diameter are those of the larger cylinder.

OR: Mean and median incomes

Data (thousand rupees): 20,22,25,90,23. Sum=180, so mean= $180/5=36$.

Ordered data: 20,22,23,25,90; median=23.

The value 90 is an extreme outlier and pulls the mean upward.

Mean = 36 thousand; median = 23 thousand. The median better represents the typical income.

Q2(ix) Verify $P\times Q = \text{LCM}\times\text{HCF}$

$P\times Q=(6x^2y)(9xy^2)=54x^3y^3$.

$\text{HCF}=3xy$ and $\text{LCM}=18x^2y^2$. Their product= $(3xy)(18x^2y^2)=54x^3y^3$.

Both sides equal $54x^3y^3$; verified.

OR: Coin relative frequencies

Heads=15, total=50, so $\text{RF}(H)=15/50=3/10$.

Tails= $50-15=35$, so $\text{RF}(T)=35/50=7/10$.

Relative frequencies: Heads=3/10; Tails=7/10.

Section C - Complete solutions (including every OR part)

Q3 Find p and q so x^4-12x^3+px+q is a complete square

Assume $x^4-12x^3+px+q=(x^2+ax+b)^2$.

Expansion gives $x^4+2ax^3+(a^2+2b)x^2+2abx+b^2$.

$2a=-12 \Rightarrow a=-6$. Since the given x^2 coefficient is 0: $a^2+2b=0 \Rightarrow 36+2b=0 \Rightarrow b=-18$.

$p=2ab=2(-6)(-18)=216$ and $q=b^2=324$.

$p=216$, $q=324$; polynomial= $(x^2-6x-18)^2$.

OR: Interior angles for A(-2,0), B(3,0), C(6,5)

$AB=5$, $BC=\sqrt{34}$ and $AC=\sqrt{89}$.

At A: $\cos A = (AB^2 + AC^2 - BC^2) / (2 \cdot AB \cdot AC) = 80 / (10\sqrt{89}) = 8/\sqrt{89}$, so $A \approx 32.01^\circ$.

At B: $\cos B = (AB^2 + BC^2 - AC^2) / (2 \cdot AB \cdot BC) = -30 / (10\sqrt{34}) = -3/\sqrt{34}$, so $B \approx 120.96^\circ$.

$C = 180^\circ - A - B \approx 27.03^\circ$.

$\angle A \approx 32.01^\circ$, $\angle B \approx 120.96^\circ$, $\angle C \approx 27.03^\circ$.

Q4 Person and tower

The line of sight, horizontal through the person's eye, and the vertical tower form a right-angled triangle. Use the tangent ratio.

If horizontal distance from the person to the tower is d feet and total tower height is H, then $\tan 30^\circ = (H-6)/d$.

Therefore $H = 6 + d \tan 30^\circ = 6 + d/\sqrt{3}$.

The printed question lacks a numerical horizontal distance, so a unique numerical height cannot be found. General result: $H = 6 + d/\sqrt{3}$ feet.

In an exam, explicitly state the missing datum; inventing a distance would be mathematically incorrect.

OR: Convert $4x-3y+9=0$ into standard forms

Slope-intercept: $-3y = -4x - 9 \Rightarrow y = (4/3)x + 3$.

Two-intercept: x-intercept $= -9/4$, y-intercept $= 3 \Rightarrow x/(-9/4) + y/3 = 1$.

Symmetric: using point $(-9/4, 0)$ and direction $(3, 4)$: $(x + 9/4)/3 = y/4$.

Normal: multiply by -1 and divide by $\sqrt{4^2 + (-3)^2} = 5$: $(-4/5)x + (3/5)y = 9/5$.

All four required forms are shown above.

Q5 Construct triangle ABC: AB=5.8 cm, BC=4.9 cm, $\angle B=60^\circ$; draw angle bisectors

1. Draw $AB=5.8$ cm.
2. At B construct a 60° ray. On it mark C so $BC=4.9$ cm. Join A to C.
3. Bisect $\angle A$: with A as centre draw an arc cutting AB and AC; from those two cut points draw equal arcs and join A to their intersection.
4. Repeat at B and C. Let the bisectors meet at I.
5. Drop perpendiculars from I to all three sides. These perpendicular distances are equal, confirming that I is the incentre.

Yes. All three internal angle bisectors are concurrent at the incentre I.

OR: Mean of grouped data

Class marks x: 145, 155, 165, 175, 185; frequencies f: 6, 9, 13, 7, 5.

$\Sigma f = 40$. Products fx: 870, 1395, 2145, 1225, 925; hence $\Sigma fx = 6560$.

Mean $= \Sigma fx / \Sigma f = 6560 / 40 = 164$.

Mean = 164.

Paper 2 - Version 12081

Section A - MCQs with detailed reasoning

MCQ 1. Evaluate $16^{1/2}$.

A. 2	B. 8	C. 4	D. 16
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Correct answer: C. $16^{1/2} = \sqrt{16} = 4$.

Why the other options are wrong: 2, 8 and 16 do not square to 16 as the principal square root requires.

MCQ 2. Correct scientific notation of 350 is:

A. 35×10^2	B. 3.5×10^2	C. 35×10^3	D. 350×10^1
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Correct answer: B. $350 = 3.5 \times 100 = 3.5 \times 10^2$, with coefficient between 1 and 10.

Why the other options are wrong: A = 3500, C = 35000 and D = 3500; their coefficients also fail normalized scientific notation.

MCQ 3. Find the domain of $\{(2,4),(3,5)\}$.

A. {2,3}	B. {4,5}	C. {3,4}	D. {2,4}
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Correct answer: A. The domain is the set of first coordinates: {2,3}.

Why the other options are wrong: B is the range; C and D mix first and second coordinates.

MCQ 4. Find the HCF of $8x^2$, $12x^3$, $24x^2y^2$.

A. $4x^2$	B. $4x$	C. $8x^2$	D. $12x^3$
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Correct answer: A. $\gcd(8,12,24)=4$ and the minimum x-power is 2; y is absent from two terms. $\text{HCF} = 4x^2$.

Why the other options are wrong: B omits a common x; C does not divide $12x^3$; D does not divide $8x^2$.

MCQ 5. $x + 5 > 9$ implies:

A. $x = 0$	B. $x > 4$	C. $x < 4$	D. $x = 4$
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Correct answer: B. Subtract 5: $x > 4$.

Why the other options are wrong: $x=0$ and $x=4$ fail the strict inequality; $x<4$ gives values below the boundary.

MCQ 6. In the right triangle, adjacent side to $\theta = 2$ and hypotenuse = 4. Find θ .

A. 0°	B. 30°	C. 60°	D. 90°
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Correct answer: C. $\cos \theta = \text{adjacent/hypotenuse} = 2/4 = 1/2$; therefore $\theta = 60^\circ$.

Why the other options are wrong: $\cos 0^\circ=1$, $\cos 30^\circ=\sqrt{3}/2$ and $\cos 90^\circ=0$.

MCQ 7. A ship moves on a bearing of 90° . Its direction is:

A. North	B. South	C. West	D. East
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Correct answer: D. 090° measured clockwise from North is due East.

Why the other options are wrong: North is 000° , South 180° and West 270° .

MCQ 8. Two polygons are similar if they have:

A. Equal area	B. Equal perimeters	C. Equal corresponding angles	D. Equal volumes
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Correct answer: C. Similar polygons have equal corresponding angles and proportional corresponding sides.

Why the other options are wrong: Their areas and perimeters need not be equal, only related by scale factors; volume does not apply to polygons.

MCQ 9. Which is a fundamental trigonometric identity?

A. $\sin \theta = \cos \theta$	B. $\sin^2 \theta + \cos^2 \theta = 1$	C. $\tan \theta = \sin \theta$	D. $\sec \theta = \cos \theta$
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Correct answer: B. The Pythagorean identity is $\sin^2 \theta + \cos^2 \theta = 1$.

Why the other options are wrong: A holds only for special angles; C misses division by $\cos \theta$; D confuses secant with the reciprocal of cosine.

MCQ 10. The angle bisectors of a triangle are concurrent at the:

A. Incentre	B. Orthocentre	C. Circumcentre	D. Centroid
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Correct answer: A. The three internal angle bisectors meet at the incentre.

Why the other options are wrong: Orthocentre: altitudes; circumcentre: perpendicular bisectors; centroid: medians.

MCQ 11. A regular polygon has exterior angle 36° . Find its number of sides.

A. 5	B. 6	C. 9	D. 10
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Correct answer: D. For a regular polygon, $n = 360^\circ / 36^\circ = 10$.

Why the other options are wrong: 5, 6 and 9 would give exterior angles 72° , 60° and 40° .

MCQ 12. The locus of points at a fixed distance from a given point is a:

A. Straight line	B. Triangle	C. Circle	D. Altitude
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Correct answer: C. A circle is the set of all points at a fixed distance (radius) from its centre.

Why the other options are wrong: The other figures do not keep every point at one fixed distance from a single point.

MCQ 13. The modal class in a grouped frequency distribution is the class with:

A. Highest frequency	B. Lowest frequency	C. Median frequency	D. Mean frequency
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Correct answer: A. By definition, the modal class has the greatest frequency.

Why the other options are wrong: Lowest, median and mean frequencies do not define the mode.

MCQ 14. If $P(E) = 3/5$, find $P(E')$.

A. $2/5$	B. $3/5$	C. $5/3$	D. $1/5$
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Correct answer: A. $P(E') = 1 - P(E) = 1 - 3/5 = 2/5$.

Why the other options are wrong: $3/5$ is the original event; $5/3$ exceeds 1; $1/5$ does not make the total 1.

MCQ 15. In a class, 18 out of 40 students received grade A. Relative frequency is:

A. $6/40$	B. $18/40$	C. $2/40$	D. $8/40$
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Correct answer: B. Relative frequency = number with grade A / total = $18/40 = 9/20$.

Why the other options are wrong: The other numerators do not equal the observed 18 successes.

Section B - Complete solutions (including every OR part)

Q2(i) Simplify the product of fourth roots

$$\sqrt[4]{(a^3/b^3)} \times \sqrt[4]{(b^3/c^3)} \times \sqrt[4]{(c^3/a^3)} = \sqrt[4]{[(a^3b^3c^3)/(b^3c^3a^3)]} = \sqrt[4]{1}.$$

Answer = 1 (for nonzero a,b,c, with real-root conditions as applicable).

OR: Verify De Morgan's law

$$U = \{1,2,3,4,5,6\}, A = \{2,5,6\}, B = \{1,2,3\}.$$

$$A \cup B = \{1,2,3,5,6\}; \text{ hence } (A \cup B)' = \{4\}.$$

$$A' = \{1,3,4\} \text{ and } B' = \{4,5,6\}; \text{ therefore } A' \cap B' = \{4\}.$$

$(A \cup B)' = A' \cap B' = \{4\}$. A Venn diagram shades only element 4 on both sides.

Q2(ii) Cyclist returns directly

After 10 km south and 24 km west, the cyclist is southwest of the start. The return vector is 24 km east and 10 km north.

$$\text{Bearing angle from North: } \tan \theta = \text{east/north} = 24/10, \text{ so } \theta = \tan^{-1}(2.4) \approx 67.38^\circ.$$

Simple bearing for the return journey $\approx 067.4^\circ$ (N67.4°E).

OR: Similar triangles with AD=5 cm, DB=3 cm, DE||BC

$$AB = 5 + 3 = 8 \text{ cm. Since } DE \parallel BC, \triangle ADE \sim \triangle ABC \text{ and linear scale} = AD/AB = 5/8.$$

$$(a) DE/BC = 5/8.$$

$$(b) \text{Area}(\triangle ADE)/\text{Area}(\triangle ABC) = (5/8)^2 = 25/64.$$

$$(c) \text{Area}(\triangle ADE) = 256 \times 25/64 = 100 \text{ cm}^2.$$

$$(d) \text{Area trapezium DBCE} = 256 - 100 = 156 \text{ cm}^2.$$

Answers: 5/8; 25/64; 100 cm²; 156 cm².

Q2(iii) Compare 8.5 and 7.8 Richter earthquakes

$$\text{Intensity ratio} = 10^{(8.5-7.8)} = 10^{0.7} \approx 5.012.$$

The 1920 earthquake was about 5.01 times as intense.

OR: Probability from a bag

$$P(\text{white}) = 3/10; P(\text{black}) = 7/10; \text{ sum} = 3/10 + 7/10 = 1.$$

Answers: 3/10, 7/10 and 1.

Q2(iv) Compound inequality

$$3x+5 \leq 5-3(x+2) \leq 6x-10. \text{ The middle expression is } -3x-1.$$

$$\text{Left: } 3x+5 \leq -3x-1 \Rightarrow 6x \leq -6 \Rightarrow x \leq -1.$$

$$\text{Right: } -3x-1 \leq 6x-10 \Rightarrow 9 \leq 9x \Rightarrow x \geq 1.$$

No real number can satisfy $x \leq -1$ and $x \geq 1$ simultaneously.

Solution set = $\emptyset$ (no solution).

OR: Factorize $(x^2+5x+4)(x^2+5x+6)-3$

$$\text{Let } t = x^2+5x. \text{ Then } (t+4)(t+6)-3 = t^2+10t+21-3 = (t+3)(t+7).$$

$(x^2+5x+3)(x^2+5x+7)$.

Q2(v) Verify radical identity

$$\text{LHS} = \sqrt{[(1-\cos\theta)/(1+\cos\theta)]}. \text{ Multiply numerator and denominator inside the radical by } (1-\cos\theta):$$

$$\text{LHS} = \sqrt{[(1-\cos\theta)^2/(1-\cos^2\theta)]} = \sqrt{[(1-\cos\theta)^2/\sin^2\theta]} = (1-\cos\theta)/\sin\theta \text{ for an acute } \theta.$$

$$\text{Rationalizing gives } (1-\cos\theta)/\sin\theta = \sin\theta/(1+\cos\theta) = \text{RHS}.$$

Verified (with the usual acute-angle/domain assumption).

OR: Regular polygon

Exterior angle = 12° , so $n = 360^\circ / 12^\circ = 30$.

Sum of interior angles = $(n-2)180^\circ = 28 \times 180^\circ = 5040^\circ$.

If each side is 1 m, perimeter = $30 \times 1 = 30$ m.

30 sides; interior-angle sum 5040°; perimeter 30 m.

Q2(vi) Perpendicular lines: find k

$$m_{AB} = (8-3)/(2-0) = 5/2. \quad m_{CD} = (k-10)/(8-3) = (k-10)/5.$$

For perpendicular lines, $m_{AB} m_{CD} = -1$: $(5/2)((k-10)/5) = -1$.

$$(k-10)/2 = -1 \Rightarrow k = 8.$$

k=8.

OR: Line through intersection and (-1,0)

Solve $3x+2y+1=0$ and $x-2y+3=0$. Adding gives $4x+4=0 \Rightarrow x=-1$; then $y=1$.

The intersection is $(-1,1)$. The second given point is $(-1,0)$; both have $x=-1$.

Required vertical line: $x=-1$, or $x+1=0$.

Q2(vii) Tower equidistant from P(2,5), Q(8,3)

$$\text{Equate distances: } (x-2)^2 + (y-5)^2 = (x-8)^2 + (y-3)^2.$$

$$\text{Expansion gives } 12x - 4y - 44 = 0 \Rightarrow 3x - y - 11 = 0.$$

Locus: $3x - y - 11 = 0$, i.e. $y = 3x - 11$.

OR: Perpendicular to $y=2x-5$ at (3,1)

Given slope = 2, so perpendicular slope = $-1/2$.

$$y-1 = (-1/2)(x-3) \Rightarrow x+2y-5=0.$$

Required line: $x+2y-5=0$.

Q2(viii) Cartesian product and R_1

$$A \times B = \{(1,2), (1,4), (3,2), (3,4)\}.$$

Condition $x > y$ is true only for $(3,2)$.

$A \times B = \{(1,2), (1,4), (3,2), (3,4)\}$; $R_1 = \{(3,2)\}$.

OR: Relative frequency in 150 tosses

$$\text{RF(Heads)} = 70/150 = 7/15.$$

$$\text{Tails} = 150 - 70 = 80; \text{RF(Tails)} = 80/150 = 8/15.$$

Heads = 7/15; Tails = 8/15.

Q2(ix) Verify product rule for $A=4a^2b$, $B=6ab^2$

$$A \times B = 24a^3b^3.$$

$$\text{HCF} = 2ab; \text{LCM} = 12a^2b^2; \text{product} = (2ab)(12a^2b^2) = 24a^3b^3.$$

$A \times B = \text{LCM} \times \text{HCF} = 24a^3b^3$.

OR: Mode, median and mean

Data: 6, 8, 6, 5, 7, 6, 9. Ordered: 5, 6, 6, 6, 7, 8, 9.

Mode = 6; median = 4th value = 6; mean = $(47)/7 \approx 6.714$.

Mode = 6, median = 6, mean = 47/7 $\approx$ 6.71.

Section C - Complete solutions (including every OR part)

Q3 Construct $\triangle ABC$: $AB=5.8$ cm, $BC=5$ cm, $\angle B=45^\circ$; draw medians

1. Draw $AB=5.8$ cm. At B construct a 45° ray; mark C on it so $BC=5$ cm. Join AC.
2. Construct the midpoint D of BC, midpoint E of CA and midpoint F of AB using perpendicular-bisector arcs.
3. Draw AD, BE and CF. Let them meet at G.
4. Measure along each median: G divides every median in the ratio 2:1 measured from the vertex ($AG:GD=BG:GE=CG:GF=2:1$).

Yes. The three medians are concurrent at centroid G.

OR: Interior angles of $X(-3,2)$, $Y(0,-1)$, $Z(3,3)$

$$XY=3\sqrt{2}, YZ=5, XZ=\sqrt{37}.$$

Using cosine rule: $\angle X \approx 54.46^\circ$, $\angle Y \approx 81.87^\circ$, and $\angle Z \approx 43.67^\circ$. Their sum is 180° .

$\angle X \approx 54.46^\circ$, $\angle Y \approx 81.87^\circ$, $\angle Z \approx 43.67^\circ$.

Q4 Tower 120 m, angles of depression 60° and 45°

Let distances of nearer and farther boats from the tower base be d_1 and d_2 .

$$\tan 60^\circ = 120/d_1 \Rightarrow d_1 = 120/\sqrt{3} = 40\sqrt{3} \text{ m.}$$

$$\tan 45^\circ = 120/d_2 \Rightarrow d_2 = 120 \text{ m.}$$

They are on the same side, so separation $= d_2 - d_1 = 120 - 40\sqrt{3}$.

Distance between boats = $120 - 40\sqrt{3} \approx 50.72$ m.

OR: Convert $x-2y+4=0$

Slope-intercept: $y = (1/2)x + 2$.

Two-intercept: x -intercept $= -4$, y -intercept $= 2 \Rightarrow x/(-4) + y/2 = 1$.

Symmetric: through $(-4,0)$ with direction $(2,1)$: $(x+4)/2 = y/1$.

Normal: multiply by -1 and divide by $\sqrt{5}$: $(-1/\sqrt{5})x + (2/\sqrt{5})y = 4/\sqrt{5}$.

All four forms are shown above.

Q5 Find k for a perfect square

$$(y^2 + 2/y^2 + 2)^2 = y^4 + 4/y^4 + 4 + 4 + 4y^2 + 8/y^2$$

$$= y^4 + 4y^2 + 8 + 8/y^2 + 4/y^4.$$

Comparing with $y^4 + 4y^2 + k + 8/y^2 + 4/y^4$ gives $k=8$.

$k=8$; expression $= (y^2 + 2/y^2 + 2)^2$.

OR: Median of grouped marks

Total $N=50$, so $N/2=25$. Cumulative frequencies: 5,15,30,42,50; median class is 21-30.

Using continuous boundary $L=20.5$, previous cumulative frequency $c=15$, class frequency $f=15$, width $h=10$:

$$\text{Median} = L + [(N/2 - c)/f]h = 20.5 + [(25 - 15)/15]10 = 27.1667.$$

Median marks ≈ 27.17 .

Paper 3 - Version 19081

Section A - MCQs with detailed reasoning

MCQ 1. Evaluate $(1/9)^{-1/2}$.

A. 3	B. 6	C. 9	D. 81
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Correct answer: A. $(1/9)^{-1/2} = 9^{1/2} = \sqrt{9} = 3$.

Why the other options are wrong: 6, 9 and 81 correspond to neither the reciprocal nor the square-root exponent.

MCQ 2. Scientific notation of 71000 is:

A. 7.1×10^2	B. 0.71×10^2	C. 7.1×10^3	D. 7.1×10^4
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Correct answer: D. Move the decimal four places left: $71000 = 7.1 \times 10^4$.

Why the other options are wrong: A=710, B=71 and C=7100.

MCQ 3. Find the range of $\{(1,3),(2,5)\}$.

A. {1,2}	B. {3,5}	C. {1,3}	D. {2,5}
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Correct answer: B. The range is the set of second coordinates: {3,5}.

Why the other options are wrong: A is the domain; C and D mix first and second coordinates.

MCQ 4. For polynomials A and B, HCF $\times$ LCM equals:

A. A+B	B. A-B	C. $(\text{HCF})^2$	D. A $\times$ B
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Correct answer: D. For the usual monomial/polynomial factorization setting, product of HCF and LCM equals product A $\times$ B (up to units/sign convention).

Why the other options are wrong: Sum, difference and HCF squared do not reproduce all maximum and minimum prime-factor powers.

MCQ 5. $-2x \leq 4$ gives:

A. $x \leq -2$	B. $x \geq -2$	C. $x \leq -1$	D. $x = -1$
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Correct answer: B. Dividing by -2 reverses the inequality: $x \geq -2$.

Why the other options are wrong: A fails to reverse the sign; C uses the wrong boundary; D gives only one value instead of the whole solution set.

MCQ 6. A right triangle has both perpendicular sides equal to 3. Find acute angle θ .

A. 45°	B. 60°	C. 90°	D. 120°
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Correct answer: A. $\tan \theta = \text{opposite/adjacent} = 3/3 = 1$, so $\theta = 45^\circ$.

Why the other options are wrong: 60° has $\tan \sqrt{3}$; 90° is the right angle; 120° is not an acute triangle angle.

MCQ 7. A cyclist travels on a bearing of 45° . This means movement toward:

A. South-East	B. North-East	C. South-West	D. North-West
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Correct answer: B. 045° is 45° clockwise from North, halfway toward East: North-East.

Why the other options are wrong: The other quadrants correspond to bearings 135° , 225° and 315° .

MCQ 8. If two triangles are equal, their corresponding sides are:

A. Equal	B. Unequal	C. Unproportional	D. Perpendicular
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Correct answer: A. Congruent triangles have equal corresponding sides.

Why the other options are wrong: B and C contradict congruence; D is an angle relation, not a side-length relation.

MCQ 9. $1 + \tan^2\theta$ equals:

A. $\sin^2\theta$	B. $\cos^2\theta$	C. $\sec^2\theta$	D. $\operatorname{cosec}^2\theta$
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Correct answer: C. Dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$ gives $\tan^2\theta + 1 = \sec^2\theta$.

Why the other options are wrong: A/B are bounded by 1 and do not match; D belongs to $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

MCQ 10. The perpendicular bisectors of the sides of a triangle meet at the:

A. Incentre	B. Centroid	C. Orthocentre	D. Circumcentre
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Correct answer: D. Perpendicular bisectors are concurrent at the circumcentre, equidistant from all vertices.

Why the other options are wrong: Incentre: angle bisectors; centroid: medians; orthocentre: altitudes.

MCQ 11. Each interior angle of a square is:

A. 90°	B. 60°	C. 45°	D. 30°
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Correct answer: A. A square has four right angles, each 90° .

Why the other options are wrong: 60° , 45° and 30° do not form the defining right angles of a square.

MCQ 12. The locus of points equidistant from two intersecting straight lines is formed by:

A. Two parallel lines	B. Perpendicular bisectors	C. Circles	D. Angle bisectors
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Correct answer: D. Points on either angle bisector of two intersecting lines have equal perpendicular distances from the lines.

Why the other options are wrong: Parallel lines, segment perpendicular bisectors and circles describe different loci.

MCQ 13. Find the median of 17, 27, 37, 47, 57, 67, 77.

A. 47	B. 17	C. 77	D. 0
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Correct answer: A. There are seven ordered values, so the median is the 4th value: 47.

Why the other options are wrong: 17 and 77 are extremes; 0 is not in the data.

MCQ 14. A fair die is rolled once. Probability of getting 2 is:

A. $1/2$	B. $1/6$	C. $1/3$	D. $1/5$
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Correct answer: B. One favorable face out of six equally likely faces gives $1/6$.

Why the other options are wrong: The other fractions would require 3, 2 or 1.2 favorable outcomes out of six.

MCQ 15. Which is a valid relative frequency?

A. -0.5	B. 1.2	C. 0.75	D. 2
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Correct answer: C. Relative frequency must lie from 0 to 1; 0.75 is valid.

Why the other options are wrong: -0.5 is negative, while 1.2 and 2 exceed 1.

Section B - Complete solutions (including every OR part)**Q2(i) Simplify**

Inside the square root: $216^{2/3} = (\sqrt[3]{216})^2 = 6^2 = 36$; $25^{1/2} = 5$; $(0.04)^{-1/2} = 1/\sqrt{0.04} = 5$.

Thus $\sqrt{[(36 \times 5)/5]} = \sqrt{36}$.

Answer = 6.

OR: Verify De Morgan's law

$U = \{a, b, c, d, e, f\}$, $A = \{a, b, c, d\}$, $B = \{a, c, e\}$.

$A \cup B = \{a, b, c, d, e\}$, so $(A \cup B)' = \{f\}$.

$A' = \{e, f\}$; $B' = \{b, d, f\}$; hence $A' \cap B' = \{f\}$.

$$(A \cup B)' = A' \cap B' = \{f\}.$$

Q2(ii) Reciprocal bearing of boats

Bearing of B from A is 250° . A reciprocal bearing differs by 180° .

$250^\circ - 180^\circ = 70^\circ$, written as three figures 070° .

$$\text{Bearing of A from B} = 070^\circ.$$

OR: Cartesian product for $A = \{1, 4, 8\}$, $B = \{1, 0\}$

$$A \times B = \{(1, 1), (1, 0), (4, 1), (4, 0), (8, 1), (8, 0)\}.$$

Each pair satisfies $x \geq y$, so R_1 is the entire Cartesian product.

$$R_1 = \{(1, 1), (1, 0), (4, 1), (4, 0), (8, 1), (8, 0)\}.$$

Q2(iii) Mean weight of 30 rice bags

Class marks: 7, 12, 17, 22, 27. Frequencies: 5, 7, 10, 4, 4.

$$\Sigma f = 30; \Sigma fx = 5(7) + 7(12) + 10(17) + 4(22) + 4(27) = 485.$$

$$\text{Mean} = 485/30 = 16.1667 \text{ kg}.$$

$$\text{Mean weight} \approx 16.17 \text{ kg}.$$

OR: Line through an intersection and parallel to $2x - 2y + 11 = 0$

Solve $2x + y + 2 = 0$ and $x - 2y + 1 = 0$. From the first $y = -2x - 2$; substitution gives $5x + 5 = 0$, so intersection $= (-1, 0)$.

A parallel line has form $2x - 2y + C = 0$. Through $(-1, 0)$: $-2 + C = 0 \Rightarrow C = 2$.

$$\text{Required line: } 2x - 2y + 2 = 0, \text{ or } x - y + 1 = 0.$$

Q2(iv) Solve both inequalities

$$2x - 3 < x + 1 \Rightarrow x < 4.$$

$$3x - 3 \geq 2x + 8 \Rightarrow x \geq 11.$$

No number can be both less than 4 and at least 11.

$$\text{Solution set} = \emptyset.$$

OR: Factorize $(x^2 - 7x + 4)(x^2 - 7x + 5) - 2$

Let $t = x^2 - 7x$. Then $(t + 4)(t + 5) - 2 = t^2 + 9t + 18 = (t + 3)(t + 6)$.

$$(x^2 - 7x + 3)(x^2 - 7x + 6).$$

Q2(v) Show $1 + \tan^2 \theta = \sec^2 \theta$

Start with $\sin^2 \theta + \cos^2 \theta = 1$. Divide every term by $\cos^2 \theta$:

$$\sin^2 \theta / \cos^2 \theta + 1 = 1 / \cos^2 \theta \Rightarrow \tan^2 \theta + 1 = \sec^2 \theta.$$

$$\text{Verified: } 1 + \tan^2 \theta = \sec^2 \theta.$$

OR: Gradient and equation through $(-4, -3)$, $(2, 9)$

$$m = (9 - (-3)) / (2 - (-4)) = 12/6 = 2.$$

$$\text{Using } (-4, -3): y + 3 = 2(x + 4) \Rightarrow y = 2x + 5.$$

$$\text{Gradient} = 2; \text{ equation } y = 2x + 5.$$

Q2(vi) Identify polygon

Sum of exterior angles of any polygon $= 360^\circ$. Given interior-angle sum is twice this, $S = 720^\circ$.

$$(n - 2)180^\circ = 720^\circ \Rightarrow n - 2 = 4 \Rightarrow n = 6.$$

A 6-sided polygon is a hexagon. With side 1 m, perimeter $= 6$ m.

Hexagon; perimeter = 6 m.

OR: Probability of perfect squares from 1 to 50

Perfect squares are 1,4,9,16,25,36,49: 7 numbers.

$P(\text{perfect square})=7/50$. $P(\text{not a perfect square})=1-7/50=43/50$.

Not perfect square=43/50; perfect square=7/50.

Q2(vii) Fountain equidistant from P(3,7), Q(9,1)

$$(x-3)^2+(y-7)^2=(x-9)^2+(y-1)^2.$$

Expansion and simplification give $12x-12y-24=0 \Rightarrow x-y-2=0$.

All positions lie on $y=x-2$.

OR: Line with y-intercept 4 and slope 1/2

Use $y=mx+c$ with $m=1/2$ and $c=4$.

$y=(1/2)x+4$, or $x-2y+8=0$.

Q2(viii) Similar cylinders with heights 3 cm and 5 cm

Linear ratio small:large=3:5.

(a) Volume ratio= $(3/5)^3=27:125$.

(b) Curved surface areas scale as square of linear ratio: $CSA_S=250(3/5)^2=90 \text{ cm}^2$.

(c) $V_L=162(5/3)^3=162 \times 125/27=750 \text{ cm}^3$.

Volume ratio 27:125; smaller CSA=90 cm²; larger volume=750 cm³.

OR: Continuous user growth

$$3000=500e^{0.08t} \Rightarrow 6=e^{0.08t}.$$

Taking natural logs: $\ln 6=0.08t \Rightarrow t=\ln 6/0.08 \approx 22.397$.

Time ≈ 22.4 years.

Q2(ix) Verify HCF-LCM product

$$A \times B = (4a^2b)(6ab^2) = 24a^3b^3.$$

HCF=2ab and LCM=12a²b²; product=24a³b³.

$A \times B = \text{LCM} \times \text{HCF}$; verified.

OR: Populations of nine towns

Ordered data: 18,22,24,26,26,26,30,32,36.

Mode=26; median=5th value=26.

Sum=240, so mean=240/9=26.666... .

Mode=26 thousand; median=26 thousand; mean $\approx$ 26.67 thousand.

Section C - Complete solutions (including every OR part)

Q3 Complete the polynomial to squares

Let $P=x^4+4x^3+10x^2+10x+5$.

$$(x^2+2x+3)^2=x^4+4x^3+10x^2+12x+9. \text{ Therefore } P+(2x+4)=(x^2+2x+3)^2.$$

$$\text{Also } (x^2+2x+2)^2=x^4+4x^3+8x^2+8x+4. \text{ Therefore } P-(2x^2+2x+1)=(x^2+2x+2)^2.$$

(a) Add $2x+4$. (b) Subtract $2x^2+2x+1$.

OR: Interior angles of X(-2,0), Y(1,-4), Z(6,6)

$$XY=5, YZ=5\sqrt{5}, XZ=10.$$

At X, vectors $XY=(3,-4)$, $XZ=(8,6)$; dot product $=24-24=0$, so $\angle X=90^\circ$.

$\cos Y=1/\sqrt{5}$, so $\angle Y \approx 63.435^\circ$. Hence $\angle Z \approx 26.565^\circ$.

$\angle X=90^\circ$, $\angle Y \approx 63.435^\circ$, $\angle Z \approx 26.565^\circ$.

Q4 Two men on opposite sides of a 15 m tree

For the man seeing 30° : $d_1=15/\tan 30^\circ=15\sqrt{3}$ m.

For the man seeing 60° : $d_2=15/\tan 60^\circ=5\sqrt{3}$ m.

They are on opposite sides, so distance between them $=d_1+d_2=20\sqrt{3}$ m.

Distance $= 20\sqrt{3}$ m ≈ 34.64 m.

OR: Convert $3x+4y-12=0$

Slope-intercept: $y=-(3/4)x+3$.

Two-intercept: $x/4+y/3=1$.

Symmetric: through $(4,0)$, direction $(4,-3)$: $(x-4)/4=y/(-3)$.

Normal: divide by 5: $(3/5)x+(4/5)y=12/5$.

All four forms are shown above.

Q5 Construct triangle of sides 2 cm, 3 cm, 4 cm; draw perpendicular bisectors

1. Draw $AB=4$ cm. With A as centre radius 3 cm and B as centre radius 2 cm, draw arcs meeting at C. Join AC and BC.

2. For each side, draw equal-radius arcs from its endpoints above and below the side; join the two arc intersections to obtain its perpendicular bisector.

3. Let the three perpendicular bisectors meet at O. Draw OA, OB and OC or measure them; $OA=OB=OC$.

Yes. The perpendicular bisectors are concurrent at circumcentre O.

OR: Mode of grouped electricity consumption

Highest frequency 12 gives modal class 25-29. Continuous lower boundary $L=24.5$; $h=5$; $f_1=12$, $f_0=8$, $f_2=2$.

Mode $=L+[(f_1-f_0)/(2f_1-f_0-f_2)]h$

$=24.5+[(12-8)/(24-8-2)]5=24.5+(4/14)5 \approx 25.93$.

Mode ≈ 25.93 units.

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